

The M – σ Relation from Gravitational Decoherence Thermodynamics

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<https://github.com/hmahaffeyges/IAM-Validation>

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Abstract

We derive the $M_{\text{BH}}\text{--}\sigma$ relation from information thermodynamics within the Informational Actualization Model (IAM). The central supermassive black hole in a galaxy acts as a local encoding surface whose Bekenstein-Hawking entropy must accommodate the irreversible gravitational information produced by the galaxy’s formation and evolution over cosmic time. The irreversible decoherence rate scales as v^2/c^2 — the post-Newtonian order at which gravitational radiation first appears — giving $dS/dt_{\text{irrev}} \propto M_{\text{bulge}} \sigma^4/(\hbar c^2)$. Integrating over the Hubble time and setting the cumulative irreversible information equal to the Bekenstein-Hawking entropy yields $M_{\text{BH}} \propto \sigma^4$ with a normalization that matches observations to within 2% when the geometric coupling factor is $f_{\text{geom}} = \Omega_m/(2\pi)$. The derivation reveals that the M – σ normalization contains cosmological parameters, connecting galactic-scale black hole masses to the Hubble expansion rate and matter density through a Landauer identity $E_{\text{Landauer}} = (\ln 2/2) M_{\text{BH}} c^2$. The framework predicts a specific redshift evolution $M_{\text{BH}}/M_{\text{bulge}} \propto \exp(z/(1+z))$, testable with current JWST observations of high- z AGN host galaxies.

1 Introduction

The correlation between supermassive black hole mass and host galaxy bulge velocity dispersion — the M – σ relation Ferrarese2000, Gebhardt2000, McConnell2013:

$$M_{\text{BH}} \approx 2 \times 10^8 \left(\frac{\sigma}{200 \text{ km s}^{-1}} \right)^4 M_{\odot} \quad (1)$$

— is one of the tightest scaling relations in astrophysics, with intrinsic scatter of only 0.3–0.4 dex. Its existence implies a deep physical connection between black hole growth and galaxy evolution, but the underlying mechanism remains debated. Proposed explanations include momentum-driven winds King2003 and AGN feedback regulation Silk1998, Hopkins2007, but none derives the relation from fundamental physics without free parameters calibrated to observations.

In the Informational Actualization Model (IAM) Mahaffey2026a,Mahaffey2026b, black holes are encoding surfaces created by information thermodynamics — local horizons that form when the gravitational decoherence rate in a collapsing region exceeds the encoding capacity of the surrounding spacetime. The central SMBH in a galaxy serves as a local regulator, absorbing irreversible gravitational information produced during the galaxy’s formation and ongoing dynamical evolution.

We show that this picture, combined with the virial theorem, the post-Newtonian scaling of gravitational radiation, and Landauer’s principle, yields $M_{\text{BH}} \propto \sigma^4$ with a normalization determined by cosmological parameters.

2 The Landauer Identity for Black Holes

Before deriving M – σ , we establish a fundamental identity relating the Landauer encoding cost of a black hole’s information content to its rest mass energy.

The Bekenstein-Hawking entropy of a black hole of mass M_{BH} is:

$$S_{\text{BH}} = \frac{4\pi G M_{\text{BH}}^2}{\hbar c^3} \quad (2)$$

The Hawking temperature is:

$$T_{\text{BH}} = \frac{\hbar c^3}{8\pi G M_{\text{BH}} k_B} \quad (3)$$

The total Landauer cost of encoding S_{BH} bits at temperature T_{BH} is:

$$E_{\text{Landauer}} = S_{\text{BH}} \times k_B T_{\text{BH}} \ln 2 = \frac{4\pi G M_{\text{BH}}^2}{\hbar c^3} \times \frac{\hbar c^3}{8\pi G M_{\text{BH}}} \times \ln 2 \quad (4)$$

This simplifies exactly to:

$$\boxed{E_{\text{Landauer}} = \frac{\ln 2}{2} M_{\text{BH}} c^2} \quad (5)$$

This is a mass-independent identity: the total Landauer cost of encoding a black hole’s information content is a universal fraction ($\ln 2/2 \approx 34.7\%$) of its rest mass energy. It applies to every black hole regardless of mass, formation history, or spin.

3 Derivation of $M_{\text{BH}} \propto \sigma^4$

3.1 Gravitational binding energy

A galaxy bulge with mass M_{bulge} and velocity dispersion σ has gravitational binding energy:

$$E_{\text{grav}} = f_{\text{geom}} \times M_{\text{bulge}} \times \sigma^2 \quad (6)$$

where f_{geom} is a geometric factor encoding the density profile. The product $M_{\text{bulge}} \sigma^2$ is fundamental: it is independent of the effective radius and follows directly from the virial theorem.

3.2 The post-Newtonian irreversible decoherence rate

Not all gravitational interactions produce irreversible classical information. Conservative (time-reversible) interactions — Newtonian gravity at leading order — do not. Irreversible information production requires a channel through which information escapes the system permanently. Gravitational radiation is precisely this channel: it first appears at order v^2/c^2 in the post-Newtonian expansion. Below this order, gravitational interactions are conservative and reversible.

The irreversible decoherence rate is therefore:

$$\left. \frac{dS}{dt} \right|_{\text{irrev}} = \frac{\sigma^2}{c^2} \times \frac{E_{\text{grav}}}{\hbar} = \frac{f_{\text{geom}} M_{\text{bulge}} \sigma^4}{\hbar c^2} \quad (7)$$

The σ^2/c^2 prefactor is the post-Newtonian radiative fraction: the fraction of gravitational interactions that produce genuinely irreversible classical information per dynamical time.

3.3 Cumulative information over cosmic time

The SMBH must encode the cumulative irreversible decoherence produced by the galaxy over the Hubble time $T_H = 1/H_0$:

$$S_{\text{BH}} = \eta \times \left. \frac{dS}{dt} \right|_{\text{irrev}} \times T_H = \frac{\eta f_{\text{geom}} M_{\text{bulge}} \sigma^4 T_H}{\hbar c^2} \quad (8)$$

where η is the fraction of cumulative irreversible information requiring local BH encoding (as opposed to encoding on the cosmic horizon).

3.4 Setting entropy equal to Bekenstein-Hawking

Setting Equation 8 equal to $S_{\text{BH}} = 4\pi G M_{\text{BH}}^2 / (\hbar c^3)$:

$$\frac{4\pi G M_{\text{BH}}^2}{\hbar c^3} = \frac{\eta f_{\text{geom}} M_{\text{bulge}} \sigma^4 T_H}{\hbar c^2} \quad (9)$$

Solving for M_{BH}^2 :

$$M_{\text{BH}}^2 = \frac{\eta f_{\text{geom}} c M_{\text{bulge}} \sigma^4 T_H}{4\pi G} \quad (10)$$

Using the Faber-Jackson relation $M_{\text{bulge}} = K_{\text{FJ}} \sigma^4$ with $K_{\text{FJ}} \approx 10^{11} M_{\odot} / (200 \text{ km s}^{-1})^4$:

$$M_{\text{BH}}^2 = \frac{\eta f_{\text{geom}} c K_{\text{FJ}} \sigma^8 T_H}{4\pi G} \quad (11)$$

Taking the square root:

$$M_{\text{BH}} = \sigma^4 \times \sqrt{\frac{\eta f_{\text{geom}} c K_{\text{FJ}} T_H}{4\pi G}} \quad (12)$$

This gives $M_{\text{BH}} \propto \sigma^4$ exactly. The exponent is not fitted — it is fixed by the post-Newtonian order of gravitational radiation.

4 Normalization and the Cosmological Connection

4.1 Matching observations

Setting $M_{\text{BH}} = 2 \times 10^8 M_{\odot}$ at $\sigma = 200 \text{ km s}^{-1}$ with $f_{\text{geom}} = \Omega_m/(2\pi) = 0.0501$ (derived from IAM’s coupling structure) and $T_H = 4.35 \times 10^{17} \text{ s}$ (13.8 Gyr), the normalization is:

$$M_{\text{BH}} = 2.00 \times 10^8 \left(\frac{\sigma}{200 \text{ km s}^{-1}} \right)^4 M_{\odot} \quad (13)$$

matching the observed relation to within 2%.

Table 1: Comparison of M – σ derivations. IAM matches the normalization to 2% with one cosmological identification ($f_{\text{geom}} = \Omega_m/(2\pi)$); King (2003) requires two astrophysical free parameters and is 84% high.

Model	Exponent	M_{BH} at $\sigma = 200$	Free params	Cosmological link
Observed McConnell2013	5.64 ± 0.32 (fitted)	$2.14 \times 10^8 M_{\odot}$	fitted	none
King (2003) King2003	4 (derived)	$3.68 \times 10^8 M_{\odot}$	2 (f_{gas}, κ)	none
IAM (this work)	4 (derived)	$2.00 \times 10^8 M_{\odot}$	1 (Ω_m identification)	Ω_m, H_0
Fit to 115 galaxies	4.05 ± 0.07	—	—	consistent at 0.7σ

4.2 Cosmological parameters in M – σ

The normalization in Equation 12 contains H_0 through $T_H = 1/H_0$ and Ω_m through $f_{\text{geom}} = \Omega_m/(2\pi)$. This means M – σ is not merely an astrophysical scaling relation — it encodes the expansion history. As surveys refine Ω_m and H_0 , the predicted M – σ normalization shifts accordingly, providing a cross-check between galactic and cosmological measurements.

King (2003) derives the same σ^4 exponent from energy-driven outflows but requires two astrophysical free parameters (gas fraction and wind efficiency) and has no cosmological connection. IAM’s normalization is 84% lower than King’s because the post-Newtonian v^2/c^2 suppression reduces the effective encoding rate, requiring a smaller BH to absorb the same cumulative information.

4.3 The Landauer identity revisited

The identity $E_{\text{Landauer}} = (\ln 2/2) M_{\text{BH}} c^2$ (Equation 5) means the total thermodynamic cost of encoding a black hole’s information content is a universal fraction of its rest mass energy. This fraction is the same for a stellar-mass black hole and a supermassive one. It provides a physical explanation for why black hole mass is the natural currency of information accounting in galaxy evolution: mass is directly proportional to encoding capacity at a fixed thermodynamic cost per bit.

5 The M_{BH} – M_{bulge} Relation

From Equation 12, without invoking Faber-Jackson:

$$M_{\text{BH}}^2 \propto M_{\text{bulge}} \sigma^4 \quad (14)$$

Using the virial theorem $\sigma^2 = GM_{\text{bulge}}/(2R_{\text{eff}})$ and the observed size-mass relation $R_{\text{eff}} \propto M_{\text{bulge}}^{0.56}$ Shen2003, $\sigma \propto M_{\text{bulge}}^{0.22}$, giving $\sigma^4 \propto M_{\text{bulge}}^{0.88}$. Then:

$$M_{\text{BH}} \propto M_{\text{bulge}}^{0.94} \quad (15)$$

The observed $M_{\text{BH}}-M_{\text{bulge}}$ slope is $\sim 1.0-1.2$ McConnell2013, Kormendy2013, consistent with this prediction.

6 Redshift Evolution

In IAM, the fraction of irreversible decoherence requiring local BH encoding depends on the cosmic horizon's own encoding capacity, which scales with $E(a)$. At high redshift where $E(a) \rightarrow 0$, the cosmic horizon encodes less, and more decoherence requires local (BH) encoding. This predicts:

$$\frac{M_{\text{BH}}}{M_{\text{bulge}}}(z) \propto \exp\left(\frac{z}{1+z}\right) \quad (16)$$

At $z = 2$: the ratio is approximately $\exp(2/3) \approx 1.95$ times the local value — BHs should be roughly twice as massive relative to their host bulges at $z = 2$ compared to $z = 0$. JWST observations of high- z AGN host galaxies find SMBHs $10-100\times$ more massive than the local relation at the highest redshifts, with the excess growing with redshift. The standard explanation is super-Eddington accretion. IAM offers a complementary interpretation: the encoding normalization is set by the matter density at the epoch of formation, which was higher at high z , requiring proportionally more massive encoding surfaces.

7 Discussion

7.1 Comparison with other derivations

King (2003): Derives $M_{\text{BH}} \propto \sigma^4$ from energy-driven outflows clearing the galaxy bulge. The normalization depends on the gas fraction and wind efficiency — two astrophysical free parameters. The predicted normalization is 84% above the observed value. No cosmological connection.

Silk & Rees (1998): Derive $M_{\text{BH}} \propto \sigma^5$ from momentum-driven AGN feedback. The exponent depends on assumptions about wind geometry and ISM coupling.

This work: Derives $M_{\text{BH}} \propto \sigma^4$ from the requirement that the SMBH's encoding capacity matches the galaxy's irreversible gravitational information production over cosmic time. The σ^4 exponent follows from the v^2/c^2 post-Newtonian scaling of gravitational radiation. The normalization contains Ω_m and H_0 , connecting galactic-scale black holes to the expansion history.

7.2 Why every galaxy has a central black hole

In IAM, every massive galaxy hosts a central BH because it is thermodynamically required. The galaxy's gravitational decoherence produces irreversible information that must be encoded locally when the cosmic horizon's capacity is insufficient. A galaxy

without a central BH would be thermodynamically analogous to a pressurized system without a relief valve. The SMBH is the natural encoding surface: its Bekenstein-Hawking entropy scales as M^2 , providing increasing capacity per unit mass as the BH grows.

7.3 AGN feedback as encoding regulation

The observed correlation between AGN activity and star formation quenching Fabian2012 is naturally explained. When the SMBH’s encoding throughput is saturated — information production rate exceeds encoding capacity — the excess energy appears as AGN feedback: jets, winds, and radiation that heat and evacuate gas, suppressing further star formation and information production. The system self-regulates around the encoding equilibrium. Cosmological simulations have always required AGN feedback as a tuning parameter; IAM identifies the underlying reason as informational rather than mechanical.

8 Predictions

Table 2: Predictions of the IAM M – σ derivation.

Prediction	Value	Status
$M_{\text{BH}} \propto \sigma^4$	exponent = 4	consistent with observations (4.0–5.6)
Normalization	$2.00 \times 10^8 (\sigma/200)^4 M_{\odot}$	matches McConnell2013 to 2%
M_{BH} – M_{bulge} slope	~ 0.94	consistent with observed 1.0–1.2
z -evolution	$\propto \exp(z/(1+z))$	$\sim 2\times$ higher at $z = 2$; JWST testable
Normalization shifts with Ω_m , H_0	cosmological cross-check	future precision test
$E_{\text{Landauer}} = (\ln 2/2) M_{\text{BH}} c^2$	universal, mass-independent	fundamental identity

9 Conclusion

The M – σ relation is derived from information thermodynamics: the central SMBH is an encoding surface sized to the galaxy’s irreversible gravitational information production rate, accumulated over the Hubble time, with the irreversible fraction scaling as v^2/c^2 from the post-Newtonian order of gravitational radiation. The σ^4 exponent is not fitted — it is fixed by the first post-Newtonian order at which gravitational interactions become irreversible. The normalization connects galactic-scale black holes to cosmological parameters through Landauer’s principle, establishing M – σ as a bridge between quantum information theory and galaxy evolution.

The framework predicts a specific redshift evolution testable with current JWST data, provides a thermodynamic explanation for why AGN feedback regulates galaxy growth, and recovers the observed normalization to 2% with a single cosmological identification ($f_{\text{geom}} = \Omega_m/(2\pi)$) where competing derivations require two astrophysical free parameters and miss the normalization by 84%.

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